

Week 1 — In-Class Exercise

Mapping the stakeholders, the money and the prescription for GLUCOREN

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Groups of 3-4 · 25 minutes working + 15 minutes debrief · nothing to submit; one group presents each part

Materials: this handout, the Week 1 slides (stakeholder map, waterfall, P&L), a calculator or notebook

Scenario

You are the new data scientist on the GLUCOREN brand team at Aspen Therapeutics. GLUCOREN is a branded, once-daily oral agent for type 2 diabetes, launched three years ago. It sells about 2.4 million normalized (30-day) prescriptions a year at a WAC of \$520 per script and posts roughly \$650M in net sales. Two things happened this quarter: (1) a competitor launched with a cardiovascular-outcomes claim on its label, and (2) the largest PBM signaled that GLUCOREN could lose its preferred (Tier 2) position on its national formulary next January unless the rebate improves. The brand lead has asked you to help her think through the stakeholder landscape before the commercial planning meeting.

Part A — Incentive map (10 minutes)

For each stakeholder, write (i) what they want from GLUCOREN and (ii) what would make them act against it. Then circle the two most important conflicts for the brand this year.

Stakeholder	What they want from GLUCOREN	What would make them act against it
Aspen Therapeutics (manufacturer)		
PBM (largest national)		
Payer / plan sponsor (employer, Part D plan)		
Endocrinologist / PCP (HCP)		
IDN / health system		
Patient		
Wholesaler		
Policymaker (CMS / HRSA / state)		

The two most important conflicts: _____ and _____

Part B — Gross-to-net worksheet (8 minutes)

Use the inputs below to build GLUCOREN's gross-to-net bridge for the year. All figures are illustrative.

Input	Value
WAC per normalized 30-day TRx	\$520
Annual normalized TRx	2,400,000
Wholesaler distribution fees & prompt-pay discounts	3% of gross
Channel mix (share of TRx) and rebate/discount rate on gross by channel	Commercial (PBM-managed): 55% of TRx, 40% rebate; Medicare Part D: 25% of TRx, 30% rebate; Medicaid: 10% of TRx, 60% rebate; 340B / other contract: 6% of TRx, 70% rebate; Cash / uninsured: 4% of TRx, 0% rebate
Copay assistance (commercial TRx only)	\$35 per commercial TRx
Returns & other	1% of gross
COGS	\$22 per TRx

Compute: (1) gross sales; (2) each deduction in \$; (3) net sales; (4) GTN %; (5) net revenue per TRx and contribution per TRx after COGS.

Then the scenario: the PBM asks for 5 more points of rebate on commercial volume (40% → 45%) to keep Tier 2. Marketing estimates that keeping Tier 2 preserves volume such that total TRx is 12% higher than the alternative. Is it worth it? Show the contribution under both cases.

Line	Your formula	\$
Gross sales		
Wholesaler fees		
Rebates — commercial		
Rebates — Medicare Part D		
Rebates — Medicaid		
Discounts — 340B / contract		
Copay assistance		
Returns & other		
Net sales		
GTN %		
Net revenue per TRx		
Contribution per TRx		

Scenario answer (base vs. 45% rebate with +12% TRx): _____

Part C — Follow one prescription (7 minutes)

A 58-year-old patient with type 2 diabetes and a high-deductible commercial plan sees an endocrinologist in March and receives a GLUCOREN prescription. Trace the six steps from tonight's slides. For each: which code

or identifier is created, which dataset captures it, and what could stop the prescription from becoming revenue.

Step	Codes / identifiers created	Dataset that captures it	What could stop it (and which project use case helps)
1. Diagnosis / visit			
2. Rx written			
3. Adjudication			
4. Dispense			
5. Claim paid			
6. Revenue			

Bonus: the same patient is 67 and on Medicare Part D. Which steps change, and why does the manufacturer's net price per script fall? (Hint: copay cards are prohibited in Medicare; Part D has its own manufacturer discount.)